

Tzu-Hsun Kao

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Professional Summary

Tzu-Hsun is a PhD student at the University of Colorado Boulder working with Dr. Robert Marshall. He is the lead for data processing on COSMO, a 6U CubeSat that aims to measure the Earth's magnetic field by vectorized rubidium scalar magnetometers. His research focuses on wave-driven particle precipitation, especially focusing on the impact of electromagnetic ion cyclotron (EMIC) waves. He had an MS in space science, specializing in ionospheric responses to solar eclipses, volcanic eruptions, and other geophysical events. Proficient in analyzing satellite and magnetometer data for both ionosphere study and geomagnetic research.

Key words: Ionosphere, Magnetosphere, Radiation Belt, Wave-particle interaction, CubeSat, Magnetometer

Education

University of Colorado Boulder, CO, USA <i>Ph.D. in Aerospace Engineering Sciences</i> Advisor: Robert Marshall	Aug. 2024 – Expect Dec 2027
National Central University, Taoyuan, Taiwan <i>M.S. in Space Science and Engineering</i> Thesis: Latitudinal Ionospheric Solar Eclipse Effect	Aug. 2021 – Jun. 2023
National Central University, Taoyuan, Taiwan <i>B.S. in Atmospheric Science</i>	Aug. 2017 – Jun. 2021

Awards and Honors

Graduate School Travel Grant , University of Colorado Boulder, USA	May 2026
Government Scholarships for Study Abroad , Ministry of Education, Taiwan	Jun. 2025
2nd Place Prize , Student Poster Competition in CEDAR Workshop, USA	Jun. 2023
Scholarship for Excellent Students in NCU, Taiwan	Mar. 2023
Scholarship for Academic Outstanding Students in NCU, Taiwan	Oct. 2021
Scholarship for Academic Outstanding Students in NCU, Taiwan	Oct. 2020
Academic Achievement Award in NCU, Taiwan	Sep. 2018

Experience

Research Assistant Department of Aerospace Engineering Sciences, CU Boulder	Aug. 2024 – present
Research Assistant Center for Astronautical Physics and Engineering (CAPE), NCU	Jun. 2023 – Aug. 2024
Research Assistant Department of Space Science and Engineering, NCU	Aug. 2021 – Jun. 2023
Intern Taiwan Space Agency (formerly the National Space Organization)	Jul. 2023 – Aug. 2023

Publications

- Wu, T. Y., Liu, J. Y., Lin, C. Y., Chao, C. K., **Kao, T. H.**, and Lee, P. H. (2025). The ionospheric plasma density hole on the 10 May 2024 Mother's day great magnetic storm. *Geophysical Research Letter*, 52(21), doi:10.1029/2025GL115956.
- Liu, T. C., Liu, J. Y., Lee, P. H., Lin, C. Y. T., **Kao, T. H.**, and Tsai, Y. L. (2025). Uplift of the ionosphere by tropospheric and thermospheric Lamb waves induced by the 2022 Tonga volcanic eruption. *Geoscience Letters*, 12(1), doi:10.1186/s40562-025-00419-0.
- Liu, J. Y., **Kao, T. H.**, Chen, Y. C., Chum, J., Chang, W. Y., Huang, B. S., Urbář, J., Hattori, K., Zhao, Y. X., and Chang, Y. C. (2025). Doppler Frequency shifts associated with the 18 September 2022 M6.8 Taitung Earthquake observed by local CW-HF Doppler sounding systems in Taiwan. *Journal of Geophysical Research: Space Physics*, 130(9), doi:10.1029/2025JA033876.
- Liu, J. Y., **Kao, T. H.**, Liu, T. C., Huang, B. S., Lee, P. H., Sun, Y. Y., Chen, C. H., Hattori, K., Lee, I. T., Su, C. L., Terng, C. T., and Huang, T. S. (2023) Magnetic field signatures of tropospheric and thermospheric Lamb modes triggered by the 15 January 2022 Tonga volcanic eruption, *Geophysical Research Letter*, 50(20), doi:10.1029/2023GL105393
- Kao, T. H.**, Lee, I. T., Fong, C. J., Liu, J. Y., and Chang, M. S. (2023) Investigation of the aberrant record from bus GPS receiver onboard FORMOSAT-7/COSMIC-2 satellite constellation in low Earth orbit, *Advances in Space Research*, 72(10), doi:10.1016/j.asr.2023.08.005

Conference Presentations and Papers (selected)

- Kao, T. H.**, Chism, C., Kaplan, O., Ellmeier, M., Knappe, S., Hughes, J., and Marshall, R. (May. 2026). In-Orbit Calibration of Vectorized Rubidium Magnetometer onboard COSMO, EGU General Assembly 2026, Vienna, Austria. [Link]
- Kao, T. H.**, Chism, C., Ellmeier, M., Knappe, S., Hughes, J., and Marshall, R. (Dec. 2025). COSMO: A 6U CubeSat Mission for Measuring the Earth's Magnetic Field by Using Vectorized Rubidium Scalars, AGU 2025, New Orleans, LA, USA. [Link]
- Kao, T. H.**, Chism, C., Ellmeier, M., and Marshall, R. (Jun. 2025). Design, Test, Calibration, and Data Processing of the Vectorized Scalar Magnetometer Onboard COSMO, CEDAR 2025, Des Moines, IA, USA. [Link]
- Kao, T. H.** and Liu, J. Y. (Dec. 2024) Ionospheric Responses of the 8 April 2024 and 21 June 2020 Solar Eclipse Using In-Situ Satellite Measurement and the CODE GIM, 2024 American Geophysical Union, Washington, DC, USA. [Link]
- Kao, T. H.** (Sep. 2023) The space weather and single event effect on commercial off-the-shelf bus GPS receivers onboard FORMOSAT-7/COSMIC-2 constellation, 2023 Conference on Weather Analysis and Forecasting, Taipei, Taiwan. [Link]
- Kao, T. H.**, Lee, I. T., Fong, C. J., Liu, J. Y., and Chang, M. S. (Jun. 2023) Study of Single Event Effects on Commercial off-the-shelf GPS Receivers Onboard FORMOSAT-7/COSMIC-2 Constellation in Low-Earth Orbit, Coupling, Energetics and Dynamics of Atmospheric Regions Program 2023 (CEDAR 2023), San Diego, CA, USA. [Link]
- Kao, T. H.** and Liu, J. Y. (Apr. 2023) Swarm, FORMOSAT-7/COSMIC-2, and ICON ionospheric observations during the annular solar eclipse on 21 June 2020, EGU General Assembly 2023, Vienna, Austria. [Link]
- Kao, T. H.** and Liu, J. Y. (Jun. 2022) Multiple Satellite Observations on the Ionosphere Response to the 21 June 2020 Annular Solar Eclipse, Taiwan Geoscience Assembly (TGA), Taipei, Taiwan.
- Kao, T. H.**, Lee, I. T., and Fong, C. J. (Nov. 2021) Aberrant Record on the Spacecraft Bus GPS Receiver on FORMOSAT-7 Mission, International Conference on Astronautics and Space Exploration in 2021 (iCASE 2021), Hsinchu, Taiwan.

Teaching Experiences

Teaching Assistant, Linear Algebra Department of Space Science and Engineering, NCU	Feb. 2023 – Jun. 2023
Teaching Assistant, Mechanics Department of Space Science and Engineering, NCU	Sep. 2022 – Jan. 2023
Teaching Assistant, Satellite Electrical System Design Department of Space Science and Engineering, NCU	Feb. 2022 – Jun. 2022

Mentoring Experiences

Max Tobiasen, Summer Program for Undergraduate Research (SPUR) College of Engineering & Applied Sciences, CU Boulder • Project: COSMO data processing and analysis • Description: Max works on maintaining the COSMO data pipeline code and developing an algorithm for detecting EMIC waves from COSMO data.	May 2026 – present
Oliver Kaplan, Discovery Learning Apprenticeship (DLA) College of Engineering & Applied Sciences, CU Boulder • Project: COSMO Database Handling and Calibration • Description: Ollie works on developing algorithms for determining the mounting quaternion and combining two quaternions from the two star trackers onboard COSMO. He is also developing a model to account for the third star tracker to increase the attitude accuracy.	Aug. 2025 – present